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Self-Oscillating Joule Thief Converters for MEMS Electromagnetic Energy Harvesting: Analytical Modeling and Experimental Validation

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***Abstract*—MEMS electromagnetic energy harvesters face a critical cold-start bottleneck: ultra-low voltage combined with high source impedance. Existing startup converters optimize for either low voltage or high impedance, rarely both. This paper addresses this challenge by adapting the self-oscillating Joule Thief topology through closed-form analytical modeling. Unlike prior semi-analytical approaches requiring iterative simulation, our framework enables simulation-free design exploration of BJT-based and NMOS-based implementations. For both implementations, the analytical model predicts output voltage within $10\,\%$ of experimental results over the measured voltage, load, and source-resistance sweeps. The NMOS version demonstrates a $50\,\mathrm{mV}$ start-up from a $300\,\Omega$ source impedance. Experimental validation with a $300\,\Omega$ electromagnetic energy harvester shows a trade-off in practice: zero-threshold NMOS starts at $0.35\,\mathrm{g}$ RMS acceleration but provides $43\,\%$ boost; low-threshold NMOS starts at $0.5\,\mathrm{g}$ but achieves $93\,\%$ boost—both outperforming TI BQ25570's $0.7\,\mathrm{g}$ startup requirement. By positioning the Joule Thief as a specialized startup circuit complementing high-efficiency main converters, this framework addresses a key barrier to self-powered MEMS deployment and high-impedance source energy harvesting.**

***Index Terms*—cold-start, electromagnetic transducer, high-impedance, Joule Thief, MEMS energy harvesting, self-oscillation**

I. Introduction

The proliferation of autonomous Internet-of-Things (IoT) sensor nodes has motivated significant research in ambient energy harvesting [1]. MEMS electromagnetic energy harvesters (MEMS-EMEHs) [2] offer micro-scale integration but face an intrinsic power management barrier (Table I): they generate low AC voltages from high-impedance sources under typical ambient vibrations. This arises from fundamental physics: miniaturization requires thin-wire coils (20–50 μm diameter) with hundreds of turns, inevitably yielding high coil resistance ($R \propto l/d^2$; l: wire length, d: diameter).

For high efficiency and low-voltage start-up, power management solutions commonly use a two-stage architecture: a cold-start circuit activates the system, after which a main converter takes over [9]. PMICs like TI BQ25570 [10] achieve high efficiency once operational but require $600\,\mathrm{mV}$ for cold-start. Among the converter topologies reviewed in [11], transformer-based (TF-based) designs are particularly suited for ultra-low voltage start-up (below $50\,\mathrm{mV}$).

TABLE I
Representative MEMS Electromagnetic Energy Harvesters

Ref.	Peak AC Voltage (at Accel.)	Freq.	Coil Resistance
[3]†	Coil A: $134\,\mathrm{mV}$@$0.06\,\mathrm{g}$	$52.1\,\mathrm{Hz}$	$100\,\Omega$
	Coil B: $214\,\mathrm{mV}$@$0.06\,\mathrm{g}$	$51.6\,\mathrm{Hz}$	$400\,\Omega$
	Coil C: $605\,\mathrm{mV}$@$0.06\,\mathrm{g}$	$53.2\,\mathrm{Hz}$	$1500\,\Omega$
[4]	$200\,\mathrm{mV}$@$1.8\,\mathrm{g}$	$300\,\mathrm{Hz}$	$100\,\Omega$
[5]	$805\,\mathrm{mV}$@$0.7\,\mathrm{g}$	$15\,\mathrm{Hz}$	$450\,\Omega$
[6]	$1700\,\mathrm{mV}$@$0.095\,\mathrm{g}$	$76.3\,\mathrm{Hz}$	$245\,\Omega$
[7]	$167\,\mathrm{mV}$@$7\,\mathrm{g}$	$143\,\mathrm{Hz}$	$128.2\,\Omega$
[8]	$140\,\mathrm{mV}$@$1\,\mathrm{g}$	$135\,\mathrm{Hz}$	$202\,\Omega$

† Voltage measured at optimal load; all others at open-circuit.

TF-based start-up circuits—cross-coupled oscillators [12], JFET-based oscillators [13], and Armstrong oscillators [14]—have been extensively studied for thermoelectric generators, but target low-impedance sources ($R_{\text{in}} < 10\,\Omega$). For high-impedance applications, $15\,\mathrm{mV}$ start-up at $1000\,\Omega$ has been demonstrated using an Armstrong oscillator with piezoelectric transformer [15]. However, its Barkhausen-based model requires numerical solution, and the piezoelectric transformer demands precise frequency matching while exhibiting load-dependent resonance drift. A Joule Thief implementation—a self-oscillating boost converter based on the blocking oscillator [16]—achieved $21\,\mathrm{mV}$ start-up from $400\,\Omega$ in $0.13\,\mu\mathrm{m}$ CMOS [17], but without a complete analytical framework.

The Joule Thief has been used in LED drivers [18], microbial fuel cells [19], and photovoltaic harvesting [20], yet a closed-form design methodology remains elusive. Early work [21] established comprehensive BJT models but required numerical solutions; piecewise-linear approximations [22] focused on transient behavior; and a recent fast-slow system approach [23] still requires simulation for key parameters. Another key limitation is that these analyses address only BJT implementations with ideal sources, covering neither NMOS variants nor high source impedance effects.

This paper addresses this gap by developing, to the best of our knowledge, the first closed-form analytical model for self-oscillating Joule Thief converters that incorporates source impedance. Explicit expressions for output voltage, oscillation frequency, and efficiency are derived through analysis of the

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Supplementary material (SPICE netlists, detailed derivations, and MATLAB scripts) is available online.

peak current conditions for both BJT-based and NMOS-based implementations, enabling simulation-free design exploration. Experimental validation with a $300\,\Omega$ electromagnetic energy harvester demonstrates the practical applicability of this topology for MEMS-EMEH cold-start.

The remainder of this paper is organized as follows: Section II presents the analytical framework; Section III describes the simulation and experimental methodology; Section IV presents model validation and start-up performance; Section V discusses trade-offs and comparisons; Section VI concludes.

II. Analysis of Self-Oscillating Converters

Since BJT-based and NMOS-based implementations share similar operating principles but differ in device physics, the analysis uses consistent notation, with subscripts distinguishing parameters and dynamic variables (e.g., i_{peak}, t_{on}, t_{off}) referring to the specific implementation. Table II summarizes the key symbols.

A. Assumptions and General Operating Principles

To develop tractable analytical models, we assume: (1) instantaneous switching transitions, neglecting the transient effects in calculating period T; (2) ideal coupled inductor with $k \approx 1$, negligible winding resistance, and operation below saturation; (3) constant diode forward voltage V_{D}; (4) negligible output voltage ripple; (5) idealized boundary condition $i_1(0) = 0$, though small negative currents may occur due to gate charging via magnetic coupling.

The Joule Thief converter achieves self-oscillation through positive feedback via magnetic coupling. Both implementations operate in quasi-boundary conduction mode with turn-on duration t_{on} and turn-off duration t_{off}. During turn-on, increasing primary current ($\mathrm{d}i_1/\mathrm{d}t > 0$) induces a voltage in the secondary winding that, due to inverse coupling, further drives the base/gate terminal, creating a positive feedback loop. Once the primary current i_1 reaches its peak i_{peak}, the induced voltage polarity reverses ($\mathrm{d}i_1/\mathrm{d}t < 0$), driving the device into cut-off. The periodic operation over a cycle $T = t_{\text{on}} + t_{\text{off}}$ is characterized by switch current i_{switch} and diode current i_{D}, with primary current $i_1(t)$ increasing from $i_1(0) = 0$ to $i_1(t_{\text{on}}) = i_{\text{peak}}$, then decreasing toward zero:

$$i_{\text{switch}}(t) = \begin{cases} i_1(t), & \text{for } 0 \le t < t_{\text{on}} \\ 0, & \text{for } t_{\text{on}} \le t < T \end{cases} \tag{1}$$

$$i_{\text{D}}(t) = \begin{cases} 0, & \text{for } 0 \le t < t_{\text{on}} \\ i_1(t), & \text{for } t_{\text{on}} \le t < T \end{cases} \tag{2}$$

For a pair of coupled inductors with a turns ratio $N_1 : N_2 = 1 : a$, perfect coupling yields the secondary inductance $L_2 = a^2 L_1$ and mutual inductance $M = aL_1$. The inverse coupling configuration gives:

$$V_{\text{L1}}(t) = L_1 \frac{\mathrm{d}i_1(t)}{\mathrm{d}t} - M \frac{\mathrm{d}i_2(t)}{\mathrm{d}t} \tag{3}$$

$$V_{\text{L2}}(t) = L_2 \frac{\mathrm{d}i_2(t)}{\mathrm{d}t} - M \frac{\mathrm{d}i_1(t)}{\mathrm{d}t} \tag{4}$$

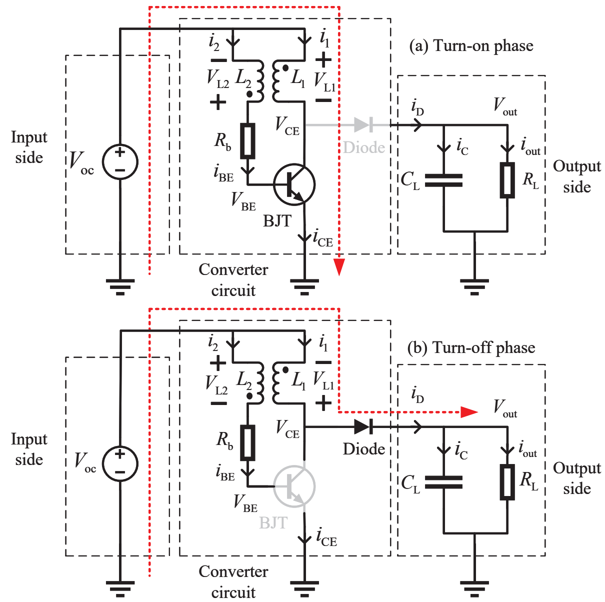

Fig. 1. Circuit configuration of the BJT-based Joule Thief circuit.

B. BJT-Based Converter Analysis

1) Circuit Configuration: Fig. 1 shows the BJT-based circuit [23], comprising a pair of coupled inductors (L_1, L_2), an NPN transistor, a base resistor R_{b}, a Schottky diode, a filter capacitor C_{L}, and a load resistor R_{L}. The circuit operates in two phases: turn-on with current through the conducting BJT [Fig. 1(a)], and turn-off with energy transfer via the diode [Fig. 1(b)]. Fig. 2 illustrates these periodic transitions.

2) Turn-on Phase: During BJT conduction, the coupled inductor feedback sustains the base-emitter voltage rise until the transistor saturates. At the peak current instant ($t = t_{\text{on}}$), both i_{CE} and V_{BE} reach their maximum values simultaneously. Applying Kirchhoff's voltage law (KVL) combined with the Ebers-Moll model yields a transcendental equation solvable via the Lambert W function (see the supplementary material for the detailed derivation). The collector-emitter voltage at peak current is:

$$V_{\text{CE}}^* = \xi_1 - V_{\text{T}} \cdot W(\xi_2) \tag{5}$$

where $W(\cdot)$ is the Lambert W function, V_{T} is the thermal voltage, and

$$\xi_1 = V_{\text{oc}} + \frac{I_{\text{s}} R_{\text{b}}}{1+a}\left(\frac{1}{\beta_{\text{F}}} + \frac{1}{\beta_{\text{R}}}\right) - \frac{V_{\text{T}}}{1+a}\left[a + \ln\left(\frac{aV_{\text{T}}\beta_{\text{R}}}{I_{\text{s}} R_{\text{b}}}\right)\right]$$

$$\xi_2 = \frac{a\beta_{\text{R}}}{(1+a)\beta_{\text{F}}} \exp\left(\frac{\xi_1}{V_{\text{T}}}\right)$$

with I_{s} the saturation current and β_{F}, β_{R} the forward/reverse current gains. The corresponding base-emitter voltage at peak current is:

$$V_{\text{BE}}^* = V_{\text{CE}}^* + V_{\text{T}} \ln\left(\frac{a\beta_{\text{R}} V_{\text{T}}}{I_{\text{s}} R_{\text{b}}}\right) \tag{6}$$

With $V_{\text{BC}}^* = V_{\text{BE}}^* - V_{\text{CE}}^*$, the peak current follows from the Ebers-Moll model:

$$i_{\text{peak}} = I_{\text{s}}\left[e^{V_{\text{BE}}^*/V_{\text{T}}} - e^{V_{\text{BC}}^*/V_{\text{T}}}\right] - \frac{I_{\text{s}}}{\beta_{\text{R}}}\left[e^{V_{\text{BC}}^*/V_{\text{T}}} - 1\right] \tag{7}$$

TABLE II
NOMENCLATURE OF KEY SYMBOLS USED IN CONVERTER ANALYSIS

Symbol	Description	Symbol	Description	Symbol	Description
V_{oc}	Open-circuit voltage	V_{BE}, V_{CE}	Base-emitter/collector-emitter voltage	β_{n}	Transconductance parameter
R_{in}	Source internal resistance	V_{CE}^*, V_{BE}^*	Voltages at peak current	W_{n}, L_{n}	Channel width/length
L_1, L_2	Primary/secondary inductance	i_{BE}, i_{CE}	Base/collector current	V_{D}	Diode forward voltage
V_{L1}, V_{L2}	Primary/secondary inductor voltage	β_{F}, β_{R}	Forward/reverse current gain	A_{d}	Diode junction area
M	Mutual inductance	I_{s}	Saturation current	V_{out}	Output voltage
k	Coupling coefficient	R_{b}	Base resistor	R_{L}	Load resistance
a	Turns ratio $\sqrt{L_2/L_1}$	V_{T}	Thermal voltage	t_{on}, t_{off}	Turn-on/turn-off duration
i_1, i_2	Primary/secondary current	V_{gs}, V_{ds}	Gate-source/drain-source voltage	T	Switching period
i_{peak}	Peak inductor current	V_{ds}^*	V_{ds} at peak current	f	Oscillating frequency
i_{switch}	Switch current	i_{ds}	Drain-source current	η	Power conversion efficiency
i_{D}	Diode current	V_{th}	Threshold voltage		

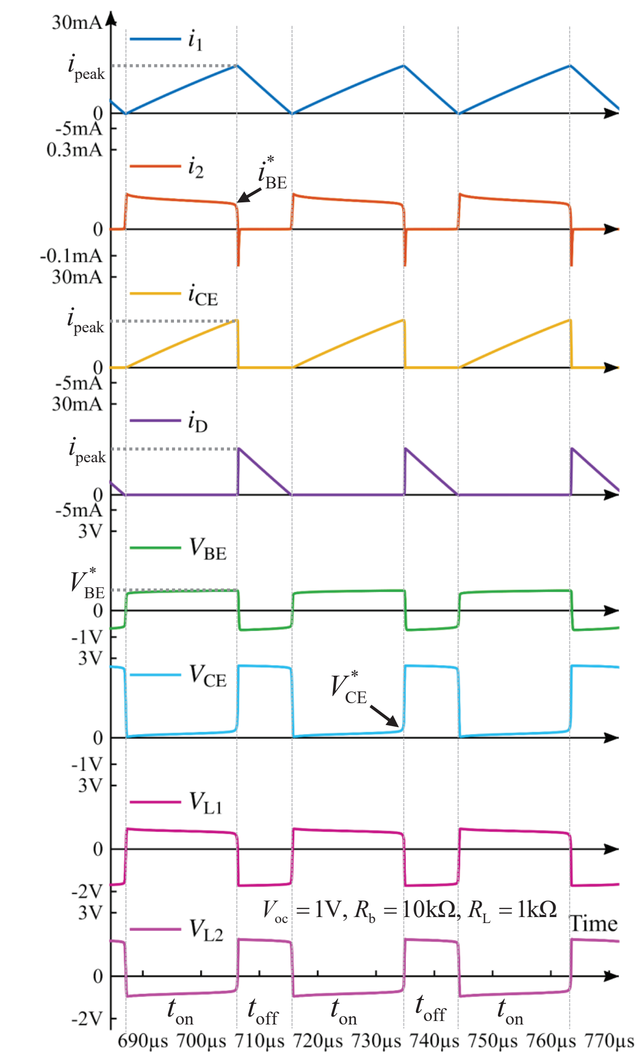

Fig. 2. Simulated steady-state waveforms (currents i_1, i_2, i_{CE}, i_{D} and voltages V_{BE}, V_{CE}, V_{L1}, V_{L2}) for the BJT-based converter, obtained from LTspice simulation.

For the turn-on phase interval, the boundary conditions are $V_{\mathrm{L1}}(0) = V_{\mathrm{oc}}$, $i_{\mathrm{CE}}(0) = 0$, $V_{\mathrm{L1}}(t_{\mathrm{on}}) = V_{\mathrm{oc}} - V_{\mathrm{CE}}^*$, and $i_{\mathrm{CE}}(t_{\mathrm{on}}) = i_{\mathrm{peak}}$. Assuming a linear $V_{\mathrm{L1}}(t)$ decrease and applying $V_{\mathrm{L1}}(t) = L_1 \cdot \mathrm{d}i_{\mathrm{CE}}(t)/\mathrm{d}t$, we can obtain:

$$t_{\mathrm{on}} = \frac{2i_{\mathrm{peak}}L_1}{2V_{\mathrm{oc}} - V_{\mathrm{CE}}^*} \tag{8}$$

3) Turn-off Phase: When the BJT enters cut-off, the primary current $i_1(t)$ redirects through the diode, decreasing linearly from i_{peak} to 0. We denote the time-average over an interval t as $\langle\cdot\rangle_t$. The inductor voltage during turn-off is:

$$\langle V_{\mathrm{L1}}\rangle_{t_{\mathrm{off}}} = -L_1\frac{i_{\mathrm{peak}}}{t_{\mathrm{off}}} \tag{9}$$

Applying KVL to the output loop and capacitor charge balance, we have:

$$V_{\mathrm{out}} = V_{\mathrm{oc}} - \langle V_{\mathrm{L1}}\rangle_{t_{\mathrm{off}}} - V_{\mathrm{D}} \tag{10}$$

$$\langle i_{\mathrm{D}}\rangle_T = \frac{i_{\mathrm{peak}}t_{\mathrm{off}}}{2(t_{\mathrm{on}} + t_{\mathrm{off}})} = \frac{V_{\mathrm{out}}}{R_{\mathrm{L}}} \tag{11}$$

Solving Eqs. (9)–(11) with t_{on} from Eq. (8) yields:

$$t_{\mathrm{off}} = \frac{i_{\mathrm{peak}}L_1(1 + \gamma_1 + \sqrt{\gamma_2})}{i_{\mathrm{peak}}R_{\mathrm{L}} + 2(V_{\mathrm{D}} - V_{\mathrm{oc}})} \tag{12}$$

$$V_{\mathrm{out}} = (V_{\mathrm{oc}} - V_{\mathrm{D}}) + \frac{i_{\mathrm{peak}}R_{\mathrm{L}} - 2(V_{\mathrm{oc}} - V_{\mathrm{D}})}{1 + \gamma_1 + \sqrt{\gamma_2}} \tag{13}$$

where the coefficients are $\gamma_1 = 2(V_{\mathrm{oc}} - V_{\mathrm{D}})/(2V_{\mathrm{oc}} - V_{\mathrm{CE}}^*)$ and $\gamma_2 = [(2V_{\mathrm{D}} - V_{\mathrm{CE}}^*)^2 + 4i_{\mathrm{peak}}R_{\mathrm{L}}(2V_{\mathrm{oc}} - V_{\mathrm{CE}}^*)]/(2V_{\mathrm{oc}} - V_{\mathrm{CE}}^*)^2$.

C. NMOS-Based Converter Analysis

1) Circuit Configuration: Fig. 3 shows the NMOS-based implementation [17]. The input is represented as a non-ideal voltage source with open-circuit voltage V_{oc} and internal resistance R_{in}. Operation includes turn-on [Fig. 3(a)] and turn-off [Fig. 3(b)] phases. Fig. 4 presents simulated steady-state waveforms.

2) Turn-on Phase: The turn-on phase analysis begins by applying KVL to the drain and gate loops of the circuit (Fig. 3). Since the gate current is negligible ($i_2 \approx 0$) during conduction, the primary current equals the drain current: $i_1 = i_{\mathrm{ds}}$. The NMOS operates in the triode region, and combining the KVL equations with the standard triode current model yields the drain current as a function of V_{ds} (see the supplementary material for the detailed derivation). At the

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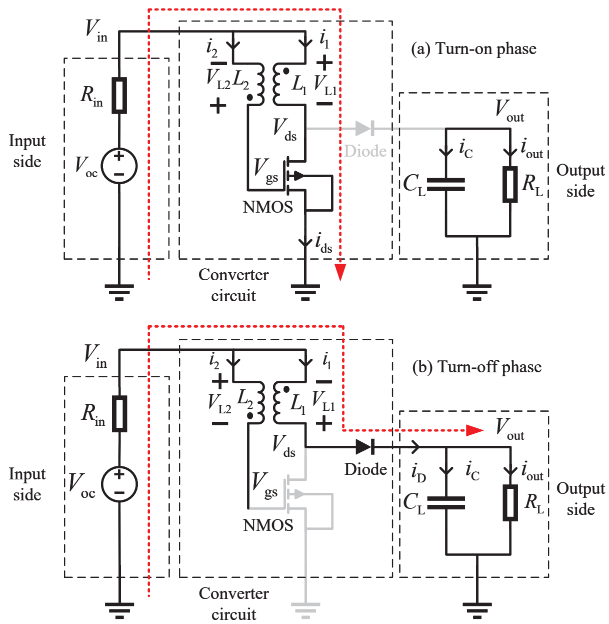

Fig. 3. Circuit configuration of the NMOS-based Joule Thief circuit.

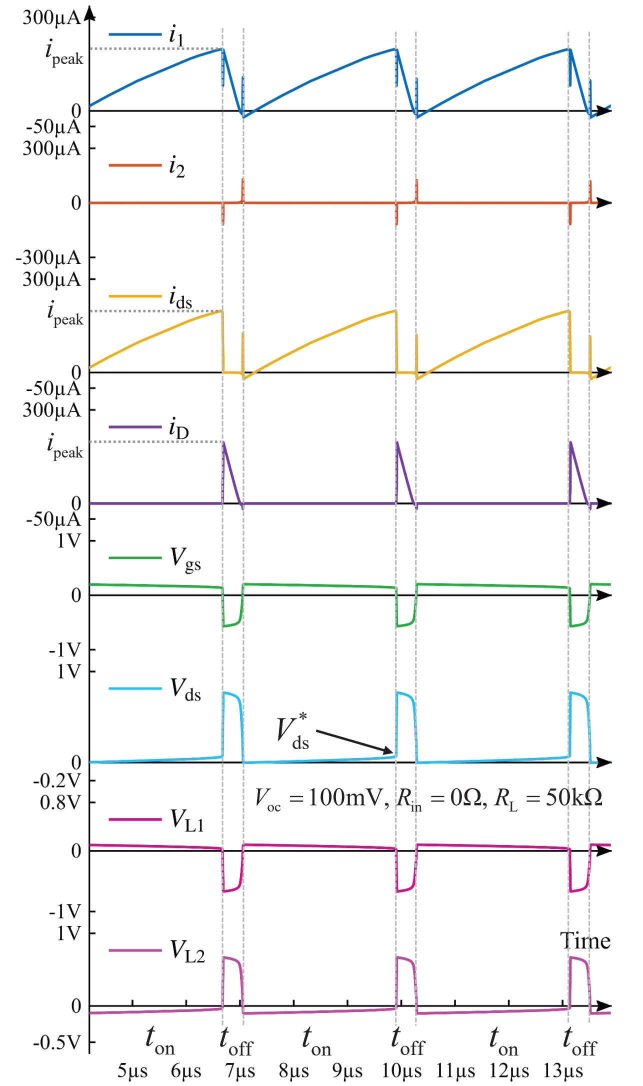

Fig. 4. Simulated steady-state waveforms (currents i_1, i_2, i_{ds}, i_{D} and voltages V_{gs}, V_{ds}, V_{L1}, V_{L2}) for the NMOS-based converter, obtained from Cadence Virtuoso simulation based on GlobalFoundries' $180\,\mathrm{nm}$ process (LV $1.8\,\mathrm{V}$ native NMOS, $L_n = 2\,\mu\mathrm{m}$, $W_n/L_n = 100$).

peak current condition ($\partial i_{\mathrm{ds}}/\partial t = 0$), applying $\partial i_{\mathrm{ds}}/\partial V_{\mathrm{ds}} = 0$ yields the closed-form expressions:

$$V_{\mathrm{ds}}^* = \frac{\sqrt{b_1(2\beta_n b_2 b_3 + b_1)} - b_1}{\beta_n b_1 b_2} \tag{14}$$

$$i_{\mathrm{peak}} = \frac{\beta_n b_2 b_3 + b_1 - \sqrt{b_1(2\beta_n b_2 b_3 + b_1)}}{\beta_n b_2^2} \tag{15}$$

where $b_1 = 1 + 2a$, $b_2 = (a+1)R_{\mathrm{in}}$, $b_3 = (a+1)V_{\mathrm{oc}} - V_{\mathrm{th}}$, and $\beta_n = \mu_n C_{\mathrm{ox}}(W_n/L_n)$ is the transconductance parameter. For an ideal source ($R_{\mathrm{in}} = 0$), these simplify to $V_{\mathrm{ds}}^* = (a+1)V_{\mathrm{oc}} - V_{\mathrm{th}}/(2a+1)$ and $i_{\mathrm{peak}} = \beta_n[(a+1)V_{\mathrm{oc}} - V_{\mathrm{th}}]^2/[2(2a+1)]$.

For the turn-on phase interval, the boundary conditions are $i_1(0) = 0$, $V_{\mathrm{L1}}(0) = V_{\mathrm{oc}}$, $i_1(t_{\mathrm{on}}) = i_{\mathrm{peak}}$, $V_{\mathrm{L1}}(t_{\mathrm{on}}) = V_{\mathrm{oc}} - R_{\mathrm{in}} i_{\mathrm{peak}} - V_{\mathrm{ds}}^*$. Assuming a linear $V_{\mathrm{L1}}(t)$ and solving $V_{\mathrm{L1}}(t) = L_1 \mathrm{d}i_1/\mathrm{d}t$, we obtain:

$$t_{\mathrm{on}} = \frac{2L_1 i_{\mathrm{peak}}}{2V_{\mathrm{oc}} - i_{\mathrm{peak}} R_{\mathrm{in}} - V_{\mathrm{ds}}^*} \tag{16}$$

3) Turn-off Phase: During turn-off, $i_1(t)$ decreases linearly from i_{peak} to zero. The average inductor voltage and the average voltage drop across R_{in} during this interval are:

$$\langle V_{\mathrm{L1}} \rangle_{t_{\mathrm{off}}} = -L_1 \frac{i_{\mathrm{peak}}}{t_{\mathrm{off}}}, \quad \langle V_{\mathrm{Rin}} \rangle_{t_{\mathrm{off}}} = R_{\mathrm{in}} \frac{i_{\mathrm{peak}}}{2} \tag{17}$$

Applying KVL and charge balance:

$$V_{\mathrm{out}} = V_{\mathrm{oc}} - \langle V_{\mathrm{L1}} \rangle_{t_{\mathrm{off}}} - \langle V_{\mathrm{Rin}} \rangle_{t_{\mathrm{off}}} - V_{\mathrm{D}} \tag{18}$$

$$\langle i_{\mathrm{D}} \rangle_T = \frac{i_{\mathrm{peak}} t_{\mathrm{off}}}{2(t_{\mathrm{on}} + t_{\mathrm{off}})} = \frac{V_{\mathrm{out}}}{R_{\mathrm{L}}} \tag{19}$$

Solving Eqs. (17)–(19) with t_{on} from Eq. (16):

$$t_{\mathrm{off}} = \frac{4 i_{\mathrm{peak}} L_1}{V_{\mathrm{ds}}^* + 2 i_{\mathrm{peak}} R_{\mathrm{in}} + 2V_{\mathrm{D}} - 4V_{\mathrm{oc}} + \sqrt{\gamma_3}} \tag{20}$$

$$V_{\mathrm{out}} = \frac{1}{4}\left(V_{\mathrm{ds}}^* - 2V_{\mathrm{D}} + \sqrt{\gamma_3}\right) \tag{21}$$

where $\gamma_3 = (V_{\mathrm{ds}}^* - 2V_{\mathrm{D}})^2 + 4R_{\mathrm{L}} i_{\mathrm{peak}}(2V_{\mathrm{oc}} - V_{\mathrm{ds}}^* - i_{\mathrm{peak}} R_{\mathrm{in}})$.

D. Key Performance Metrics and Calculation Procedure

For both implementations, the steady-state switching frequency is $f = 1/(t_{\mathrm{on}} + t_{\mathrm{off}})$, an approximation since neglected parasitic capacitances lower the actual frequency. With $P_{\mathrm{out}} = V_{\mathrm{out}}^2/R_{\mathrm{L}}$, the power conversion efficiency is:

$$\eta = \frac{P_{\mathrm{out}}}{P_{\mathrm{in}}} = \frac{V_{\mathrm{out}}^2/R_{\mathrm{L}}}{(V_{\mathrm{oc}} - \langle i_{\mathrm{in}} \rangle_T R_{\mathrm{in}}) \langle i_{\mathrm{in}} \rangle_T} \tag{22}$$

where $\langle i_{\mathrm{in}} \rangle_T$ is the input current averaged over a period T. The quasi-boundary conduction mode produces a triangular i_1 waveform (Figs. 2 and 4), yielding $\langle i_{\mathrm{in}} \rangle_T \approx \langle i_1 \rangle_T = i_{\mathrm{peak}}/2$ when neglecting secondary/gate currents.

The calculation procedure follows: (1) Define circuit parameters (source, switching device, passive components); (2) Calculate characteristic voltages and i_{peak}; (3) Determine t_{on}

TABLE III
PARAMETERS OF DISCRETE COMPONENTS USED IN THE EXPERIMENT

Component	Parameter	Value
SMMBT3904LT1G	I_s, β_F, β_R	$20.5\,\text{fA}$, 100, 1.88
ALD212900[a]	V_{th}, β_n	$0\,\text{V}$, $0.015\,\text{A}\,\text{V}^{-2}$
SiUD412ED[b]	V_{th}, β_n	$0.35\,\text{V}$, $0.9\,\text{A}\,\text{V}^{-2}$
MSD1278H-105KED	L_1, a, k, R_{ind}	$1\,\text{mH}$, 1, 0.99, $2.35\,\Omega$
RB521S30T1G	V_D	$0.3\,\text{V}$
CL31A226KAHNNNE	C_{buf}, C_L	$22\,\mu\text{F}$, $22\,\mu\text{F}$

[a] $\Delta V_{th}/\Delta T = -1.7\,\text{mV}\,{}^\circ\text{C}^{-1}$ (typ.) at $I_D = 20\,\mu\text{A}$, $V_{DS} = 0.1\,\text{V}$.
[b] $\Delta V_{th}/\Delta T = -1\,\text{mV}\,{}^\circ\text{C}^{-1}$ (typ.) at $I_D = 250\,\mu\text{A}$.

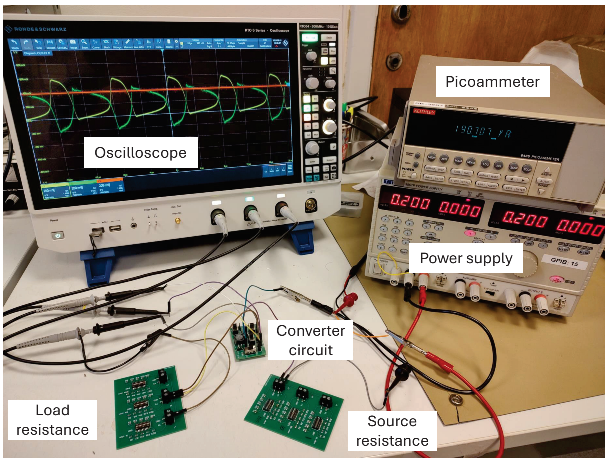

Fig. 5. Experimental setup for characterizing the self-oscillating Joule Thief converter performance.

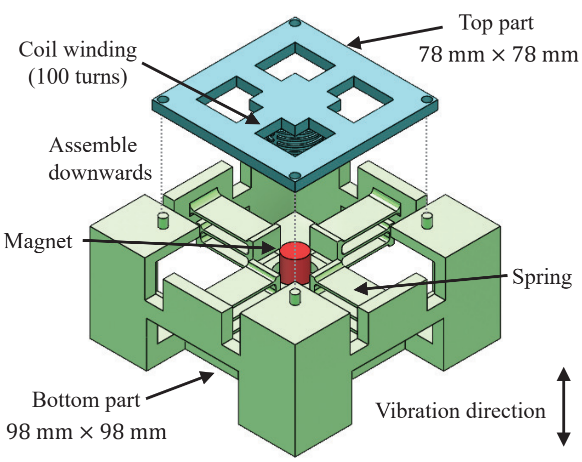

Fig. 6. Exploded view of the electromagnetic energy harvester structure: the top part ($78\,\text{mm} \times 78\,\text{mm}$) contains the 100-turn coil winding, and the bottom part ($98\,\text{mm} \times 98\,\text{mm}$) supports the spring-suspended permanent magnet that vibrates vertically.

and t_{off}; (4) Calculate V_{out}, f, and η. MATLAB scripts implementing this procedure for both BJT and NMOS converters are provided in the supplementary material.

III. SIMULATION AND EXPERIMENTAL METHODOLOGY

A. Simulation Methodology

The BJT-based converter was simulated in LTspice using the OnSemi SMMBT3904LT1G Gummel-Poon and

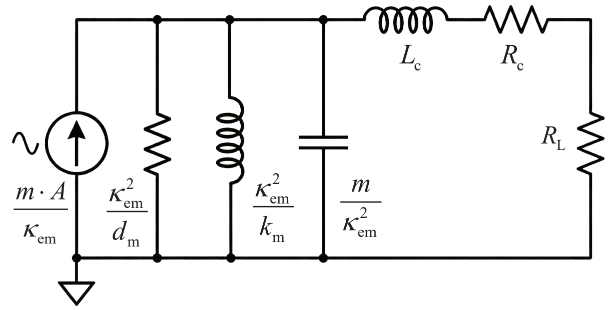

Fig. 7. Electromechanical equivalent circuit of the electromagnetic energy harvester. The mechanical domain (vibration acceleration A, mass m, stiffness k_m, damping d_m) is transformed to electrical equivalents via the electromagnetic coupling coefficient κ_{em}.

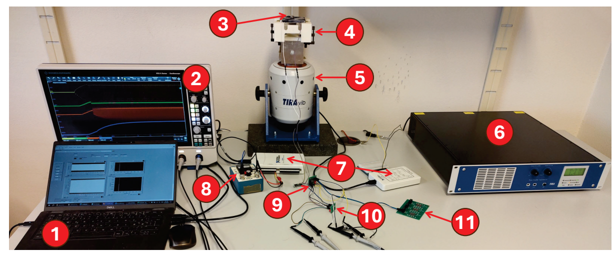

Fig. 8. Experimental setup for testing with an electromagnetic energy harvester. The platform consists of: 1) Laptop, 2) Oscilloscope, 3) Accelerometer, 4) Electromagnetic energy harvester, 5) Vibration shaker, 6) Amplifier, 7) Data acquisition devices, 8) Signal conditioner, 9) DC/DC converter, 10) AC/DC rectifier, and 11) Load resistance.

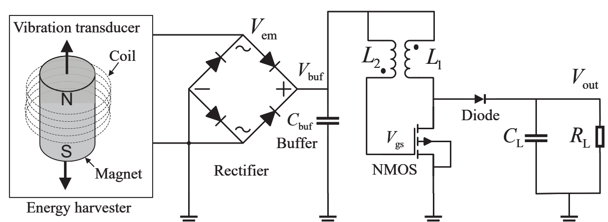

Fig. 9. Schematic of the power management interface for the electromagnetic energy harvester.

RB521S30T1G Schottky diode models, while the NMOS-based converter was simulated in Cadence Virtuoso using BSIM4 models from GlobalFoundries' $180\,\text{nm}$ process (LV $1.8\,\text{V}$ native NMOS). Both simulations used a coupled inductor model matching the Coilcraft MSD1278H-105KED ($1\,\text{mH}$, $k = 0.99$, $2.35\,\Omega$ DCR), with ideal passive components. The LTspice and Cadence Spectre netlists are provided in the supplementary material.

B. Laboratory Characterization of Discrete Prototypes

Custom PCBs with discrete components (Table III) were fabricated, and all experiments were conducted at $25\,{}^\circ\text{C}$. As shown in Fig. 5, the setup included an AIM-TTI QL355TP power supply, a Keithley 6485 picoammeter (resolution $10\,\text{fA}$, $\pm 0.1\,\%$ accuracy), a Rohde & Schwarz RTO64 oscilloscope ($10\,\text{GHz}$ bandwidth, $\pm 2\,\%$ DC gain accuracy), and switchable resistor arrays ($1\,\%$ tolerance, $10\,\Omega$–$1\,\text{M}\Omega$) for parametric sweeps. The BJT-based converter used an SMMBT3904LT1G

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TABLE IV
PARAMETERS OF THE ELECTROMAGNETIC ENERGY HARVESTER

Parameter	Symbol	Value
Proof Mass	m	$0.04\,\mathrm{kg}$
Stiffness	k_m	$4588\,\mathrm{N\,m^{-1}}$
Mechanical Damping	d_m	$0.49\,\mathrm{N\,s\,m^{-1}}$
Resonant Frequency	f_r	$53.9\,\mathrm{Hz}$
Coil Inductance	L_c	$310\,\mathrm{\mu H}$
Coil Resistance	R_c	$275\,\Omega$
Electromagnetic Coupling Coefficient	κ_em	$0.95\,\mathrm{Wb\,m^{-1}}$

with I_s and β_R from the vendor SPICE model and $\beta_\mathrm{F} = 100$ calibrated from datasheet [24] and measurements. The NMOS-based version employed an ALD212900 zero-threshold device, with V_th and β_n similarly calibrated experimentally. All converters used the Coilcraft MSD1278H-105KED (Table III). Steady-state measurements were recorded after $5\,\mathrm{s}$. Preliminary testing confirmed less than $2\,\%$ run-to-run variation across five or more trials.

C. Energy Harvester Integration Experiments

1) Harvester Design and Characterization: To validate converter performance under realistic conditions, a custom macroscale EMEH was built as an electrical emulator of MEMS-EMEH characteristics, since converter behavior largely depends on the source's electrical parameters—ultra-low voltage with high impedance—rather than transducer physical scale. The harvester employs a spring-mass-damper system, as displayed in Fig. 6. Both parts were 3D-printed using JLC temp resin [25]: the top houses the coil, while the bottom contains the spring-suspended permanent magnet assembly (ten Eclipse N814 NdFeB magnets, $4\,\mathrm{g}$ each). The coil (100 turns of $25\,\mathrm{\mu m}$-diameter copper wire) yields $R_\mathrm{c} = 275\,\Omega$ and $L_\mathrm{c} = 310\,\mathrm{\mu H}$, measured using an HP8753A network analyzer at $10\,\mathrm{kHz}$. Mechanical parameters (k_m, d_m) were extracted from open-circuit vibration testing, and κ_em was determined from the equivalent circuit model [26], as shown in Fig. 7. All parameters are listed in Table IV.

2) Experimental Setup and Test Configurations: As shown in Fig. 8, the setup employed a TIRA TV 51110 electrodynamic shaker with BAA120 amplifier for excitation at $54\,\mathrm{Hz}$ (harvester resonance). System monitoring included a TLD352A56 accelerometer with 480E09 signal conditioner and DAQ devices (NI-USB-6211, NI myDAQ). RMS acceleration levels were varied from 0.1–$0.8\,\mathrm{g}$.

Fig. 9 displays the power-conditioning circuit: a full-bridge rectifier (four RB521S30T1G Schottky diodes) with a $22\,\mathrm{\mu F}$ buffer capacitor cascaded with the DC/DC converter. Four configurations were compared: (1) Rectifier-only baseline: direct rectification without DC/DC conversion; (2) Zero Threshold Voltage NMOS Converter (ZTVNC): ALD212900 for ultra-low voltage start-up; (3) Low Threshold Voltage NMOS Converter (LTVNC): SiUD412ED with higher transconductance; (4) Commercial benchmark: TI BQ25570EVM-206 with MPPT at $50\,\%$ and buck converter disabled for fair boost-only comparison.

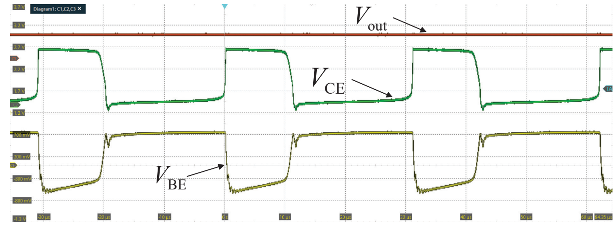

Fig. 10. Measured waveforms for the BJT-based converter under standard test conditions ($V_\mathrm{oc} = 1\,\mathrm{V}$, $R_\mathrm{b} = 10\,\mathrm{k\Omega}$, and $R_\mathrm{L} = 1\,\mathrm{k\Omega}$), showing the oscillating behavior with distinct on-off switching phases. Oscilloscope settings: $10\,\mathrm{\mu s}$/div horizontal, $500\,\mathrm{mV}$/div vertical.

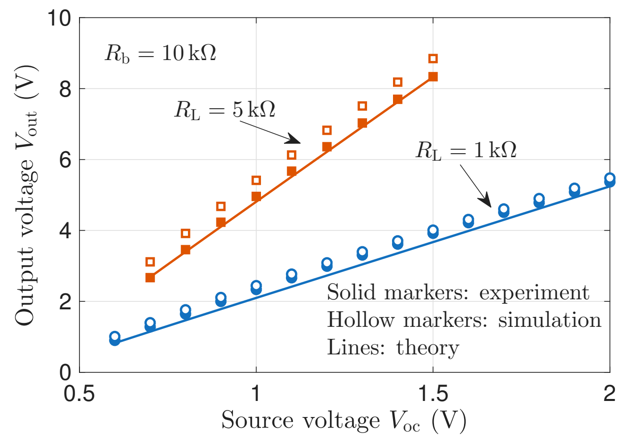

Fig. 11. Measured output voltage versus source voltage for the BJT-based implementation.

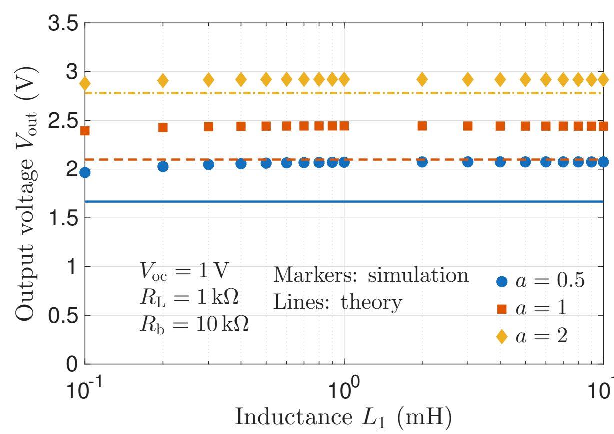

Fig. 12. Simulated output voltage versus the primary inductance and turns ratio for the BJT-based implementation.

IV. RESULTS

A. Analytical Model Validation with Laboratory Prototypes

1) BJT-Based Converter Validation: The BJT-based converter validates the analytical methodology. Measured waveforms in Fig. 10 confirm the steady-state self-oscillation, consistent with simulations in Fig. 2. As shown in Fig. 11, the analytical prediction of V_out agrees with measurements within $10\,\%$ across tested source-voltage and load conditions, with simulation tracking the same trend. The minimum operational voltage is $0.6\,\mathrm{V}$ for $R_\mathrm{L} = 1\,\mathrm{k\Omega}$. Further parametric studies in Fig. 12 validate key design insights: V_out is largely insensitive

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TABLE V
MODEL PREDICTION ERROR UNDER NON-IDEAL PARAMETERS

Parameter	Efficiency η [%] (*Err.*)[†]	Voltage V_{out} [mV] (*Err.*)[†]	Freq. f [kHz] (*Err.*)[†]
Nominal Baseline[‡] (k=0.99, $W_{\mathrm{n}}/L_{\mathrm{n}} = 500$, $A_{\mathrm{d}} = 3720\,\mu\mathrm{m}^2$)			
—	50.3 (+3.7%)	427.4 (−1.2%)	182.9 (−1.2%)
Coupling Coefficient (k)			
0.91	46.4 (−4.4%)	411.3 (−5.1%)	181.7 (−1.9%)
0.93	47.5 (−1.9%)	416.4 (−3.8%)	181.8 (−1.8%)
0.95	48.5 (+0.1%)	420.4 (−2.8%)	181.9 (−1.8%)
0.97	49.3 (+1.7%)	423.6 (−2.1%)	182.0 (−1.7%)
NMOS Aspect Ratio[§] ($W_{\mathrm{n}}/L_{\mathrm{n}}$)			
−40%	48.8 (+6.4%)	405.5 (+2.7%)	218.2 (−2.1%)
−20%	49.7 (+4.9%)	418.8 (+0.5%)	197.1 (−1.4%)
+20%	50.6 (+2.5%)	432.8 (−2.6%)	171.8 (−1.6%)
+40%	50.9 (+1.5%)	437.0 (−3.8%)	163.6 (−2.0%)
Diode Area (A_{d})			
−40%	49.1 (+1.4%)	422.2 (−2.4%)	183.5 (−0.9%)
−20%	49.6 (+2.5%)	425.1 (−1.7%)	183.0 (−1.2%)
+20%	50.7 (+4.5%)	429.0 (−0.8%)	182.5 (−1.4%)
+40%	51.2 (+5.4%)	430.7 (−0.4%)	182.3 (−1.5%)

[†] Error = (Sim. − Calc.)/Sim. × 100 %.
[‡] Assumptions in theory: ideal coupling ($k = 1$), instantaneous switching, and constant V_{D}. Simulation parameters: $V_{\mathrm{oc}} = 0.2\,\mathrm{V}$, $R_{\mathrm{in}} = 200\,\Omega$, $L_1 = 1\,\mathrm{mH}$, $a = 1$, $V_{\mathrm{th}} = 0\,\mathrm{V}$, $L_{\mathrm{n}} = 1\,\mu\mathrm{m}$, $R_{\mathrm{L}} = 10\,\mathrm{k}\Omega$.
[§] $\beta_{\mathrm{n}} = 0.086\,\mathrm{A\,V^{-2}}$ in the baseline, scaling proportionally with $W_{\mathrm{n}}/L_{\mathrm{n}}$.

TABLE VI
IMPACT OF INITIAL CURRENT ASSUMPTION ON ACCURACY

R_{L} [kΩ]	V_{out} [mV] Sim. (Err.)[†]	i_{peak} [μA] Sim. (Err.)[†]	$i_1(0)$ [μA]
10	427.4 (−1.2%)	527.7 (+1.0%)	−6.4
20	643.0 (−2.0%)	527.7 (+1.0%)	−15.5
30	807.6 (−2.6%)	527.7 (+1.0%)	−20.5
40	944.0 (−3.3%)	527.7 (+1.0%)	−26.3
50	1061.7 (−4.0%)	527.7 (+1.0%)	−31.0

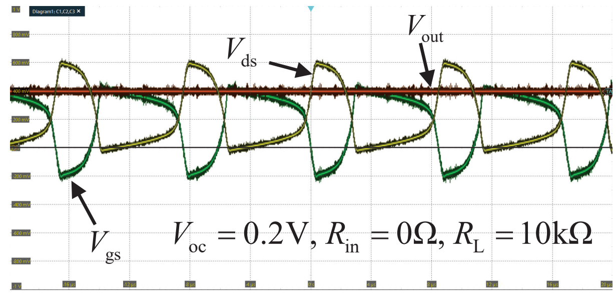
[†] Simulation parameters are identical to the nominal baseline in Table V.

(<6 % variation) to L_1 but scales with the turns ratio a. Deviations up to 20% are discussed in Section V.

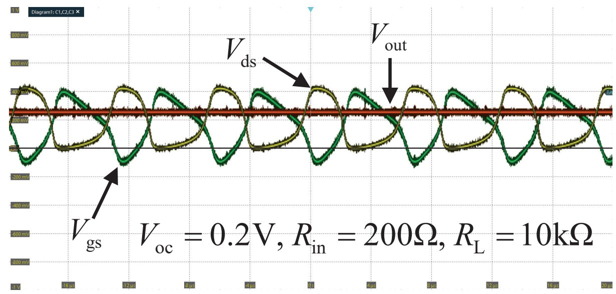
2) NMOS-Based Converter for High-Impedance Sources: Systematic sensitivity analyses in Table V quantify how key non-idealities affect model accuracy, examining coupling coefficient k, NMOS aspect ratio $W_{\mathrm{n}}/L_{\mathrm{n}}$, and diode area A_{d}. Across all parameter sweeps, the model predicts η within ±7% and V_{out} within ±6% of Cadence Spectre results, consistently overestimating frequency by 1–2%. Table VI further investigates the impact of the $i_1(0) = 0$ boundary condition by sweeping R_{L} from 10 kΩ to 50 kΩ. As R_{L} increases, the simulated initial current $i_1(0)$ becomes increasingly negative (from −6 μA to −31 μA), and the V_{out} prediction error correspondingly grows from −1.2% to −4.0%. In contrast, the i_{peak} prediction error remains constant at approximately +1% throughout the sweep.

Fig. 13 shows measured waveforms revealing R_{in}-dependent behavior: the zero-resistance case (a) features a longer period (∼8.5 μs) with rapid switching, while the 200 Ω case (b) exhibits a shorter period (∼6.5 μs) but slower transitions due to the larger RC time constant. Fig. 14 validates the model's accuracy for high-impedance sources. The converter achieves 50 mV start-up with $R_{\mathrm{in}} \geq 100\,\Omega$, and theory predicts output voltage within 10 % across all tested R_{in} values (0–300 Ω), confirming the framework's utility for design optimization.

The efficiency measurements in Fig. 15 show 38.3 % at



(a) $R_{\mathrm{in}} = 0\,\Omega$



(b) $R_{\mathrm{in}} = 200\,\Omega$

Fig. 13. Measured waveforms for the NMOS-based implementation under different source resistance conditions. Oscilloscope settings: 4 μs/div horizontal, 200 mV/div vertical.

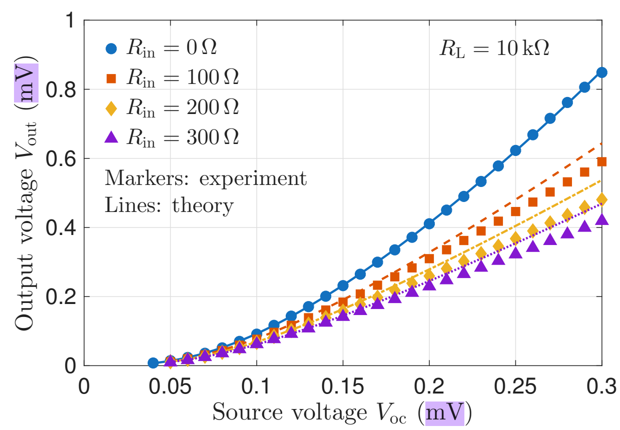


Fig. 14. Measured output voltage versus input voltage for various source resistances in the NMOS-based implementation.

$V_{\mathrm{oc}} = 300\,\mathrm{mV}$ with $R_{\mathrm{in}} = 200\,\Omega$, which is modest but sufficient for cold-start applications. At moderate R_{L} values, theory and experiment agree well. However, the discrepancy increases at higher R_{L}, consistent with the $i_1(0)$ analysis in Table VI. The underlying mechanism is discussed in Section V.

B. Energy Harvester Integration Results

1) Frequency Response and Impedance Characteristics: The harvester's open-circuit frequency response in Fig. 16 exhibits a resonant peak near 54 Hz. The equivalent circuit model, fitted at 53.9 Hz, shows close agreement with measurements across all tested acceleration levels. At higher excitation ($a_{\mathrm{rms}} \geq 0.6\,\mathrm{g}$), the experimental resonant peak shifts slightly leftward (to 53.8 Hz at $a_{\mathrm{rms}} = 0.8\,\mathrm{g}$) due to the softening effect, which reduces effective spring stiffness but is not captured by the linear equivalent circuit with constant k_{m}.

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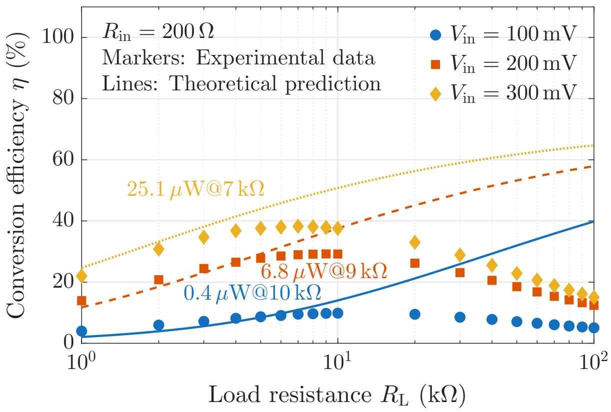

Fig. 15. Power conversion efficiency versus load resistance at different input voltages for the NMOS-based implementation (the text annotation indicates the output power and optimal resistance at maximum conversion efficiency.).

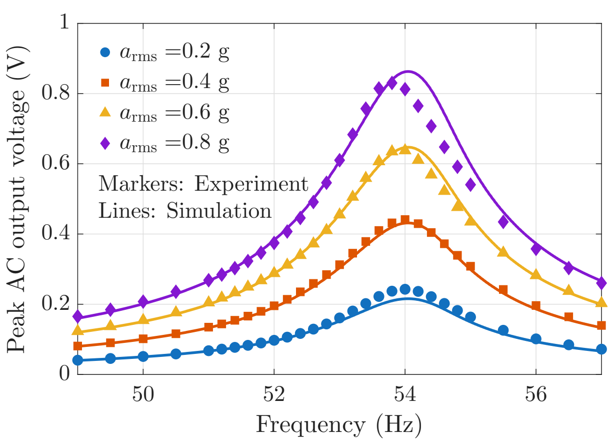

Fig. 16. Peak AC output voltage frequency response of the electromagnetic energy harvester under different acceleration levels at open-circuit conditions. Lines: equivalent circuit simulation; markers: experimental measurements.

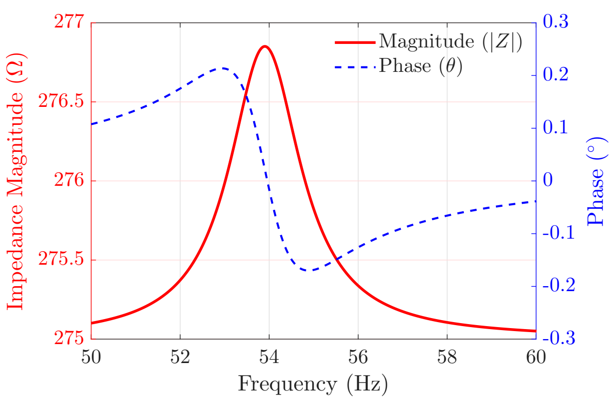

Fig. 17. Simulated impedance magnitude and phase versus frequency for the EMEH equivalent circuit.

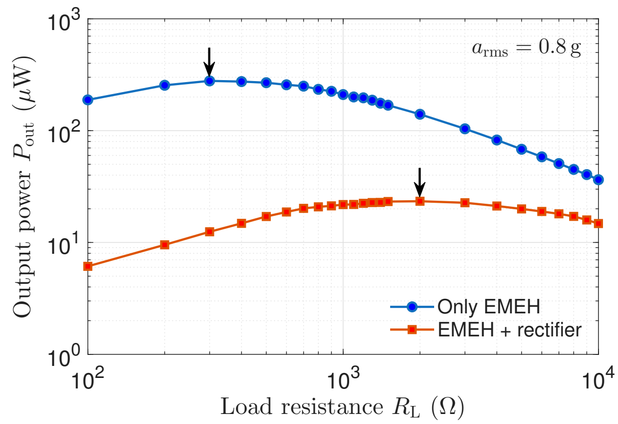

Fig. 18. Output power versus load resistance for the electromagnetic energy harvester: direct connection and with rectifier integration ($a_{\mathrm{rms}} = 0.8\,\mathrm{g}$). The arrows indicate the optimal load resistance.

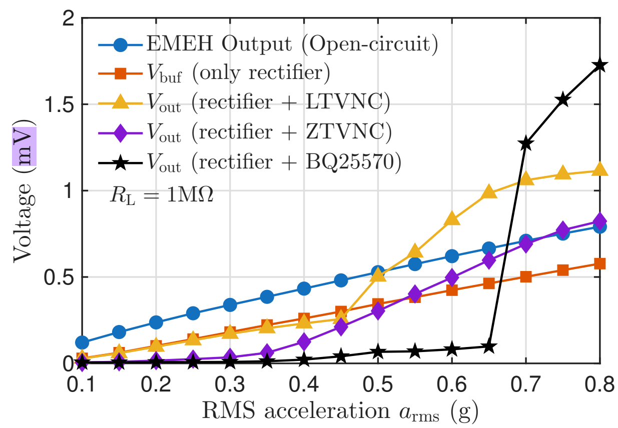

Fig. 19. Voltage characteristics versus RMS acceleration for different power conditioning stages: direct EMEH output, rectified voltage, and boosted output from two NMOS converter variants.

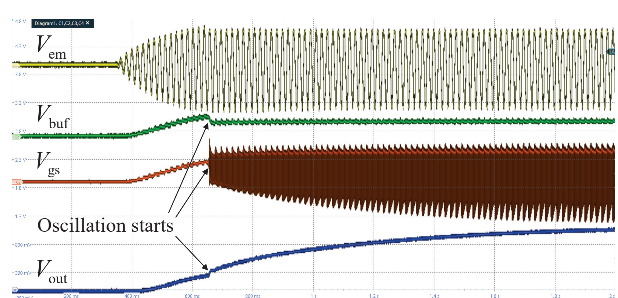

Fig. 20. Measured waveforms during the start-up process with LTVNC ($R_{\mathrm{L}} = 1\,\mathrm{M}\Omega$). Oscilloscope settings: $200\,\mathrm{ms}$/div horizontal, $500\,\mathrm{mV}$/div vertical.

The harvester's impedance characteristics are critical for power management design. As shown in Fig. 17, the simulated impedance magnitude peaks at $276.8\,\Omega$ at resonance ($53.9\,\mathrm{Hz}$) with phase crossing 0°. Experimentally, as shown in Fig. 18, the direct EMEH connection achieves maximum power $278\,\mu\mathrm{W}$ at $300\,\Omega$ load resistance. This matches both the simulated value and the theoretical optimal load $R_{\mathrm{opt}} = R_{\mathrm{c}} + \kappa_{\mathrm{em}}^2/d_{\mathrm{m}} = 276.8\,\Omega$ derived from the electromechanical model [26]. After rectifier integration, the optimal load shifts dramatically to $2\,\mathrm{k}\Omega$ with peak power reduced to $22\,\mu\mathrm{W}$.

2) Start-up Characteristics and Voltage Boost Performance: Fig. 19 shows voltage scaling across the power conditioning stages versus various excitation levels, where load resistance $R_{\mathrm{L}} = 1\,\mathrm{M}\Omega$ is chosen to maximize voltage for cold-start

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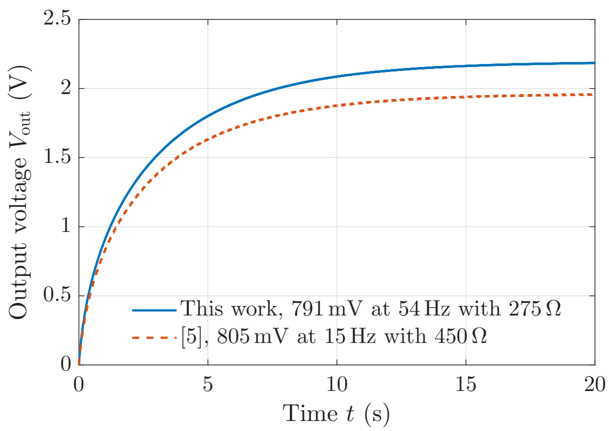

Fig. 21. Simulated output voltage for two MEMS-EMEHs modeled as AC voltage sources with a full-bridge rectifier and NMOS-based Joule Thief converter (baseline design in Table V, $C_L = 10\,\mu\text{F}$, $R_L = 1\,\text{M}\Omega$).

characterization. The direct EMEH output V_{em} scales linearly with acceleration, reaching $791\,\text{mV}$ peak (AC) at $a_{rms} = 0.8\,\text{g}$. Post-rectification, the buffer capacitor voltage V_{buf} stabilizes at $577\,\text{mV}$, representing the voltage available to the converter.

Start-up performance reveals critical differences: ZTVNC initiates oscillation at $0.35\,\text{g}$, enabled by its near-zero threshold, while LTVNC requires $0.5\,\text{g}$ due to its higher threshold but delivers significantly higher output voltage once operating. At maximum excitation ($a_{rms} = 0.8\,\text{g}$), LTVNC produces $1114\,\text{mV}$—a $93\,\%$ voltage boost from the $577\,\text{mV}$ rectified input—compared to $823\,\text{mV}$ from ZTVNC ($43\,\%$ boost). The commercial TI BQ25570 evaluation board provides an important benchmark. At maximum excitation, it delivers $1727\,\text{mV}$—the highest voltage among all configurations. However, its performance degrades dramatically at low excitation: it produces $<100\,\text{mV}$ for $a_{rms} < 0.65\,\text{g}$ because it fails to achieve cold-start.

3) Dynamic Start-up Behavior: The transient start-up process in Fig. 20 reveals the converter's dynamic behavior under vibration conditions. Successful startup is defined as sustained self-oscillation for at least 10 cycles with continuous V_{out} rise. Sinusoidal excitation at $54\,\text{Hz}$ generates the harvester AC output V_{em}. The buffer capacitor gradually accumulates charge, causing V_{buf} to rise over multiple cycles. Once V_{buf} crosses the start-up threshold, V_{gs} initiates self-oscillation (marked by arrow), resulting in a rapid increase in V_{out}. To further validate circuit applicability, Fig. 21 compares output voltage transients with [5]. Despite higher coil resistance ($450\,\Omega$ vs. $275\,\Omega$), both EMEHs reach steady-state, confirming converter robustness.

V. Discussion

A. Positioning as a Specialized Start-up Circuit

The converter is optimally positioned as a dedicated cold-start front-end within a two-stage architecture [9], complementing—rather than competing with—high-efficiency PMICs. Its strengths include ultra-low voltage operation ($50\,\text{mV}$) with high source impedance ($300\,\Omega$), self-oscillation without external drive circuitry, zero quiescent current in

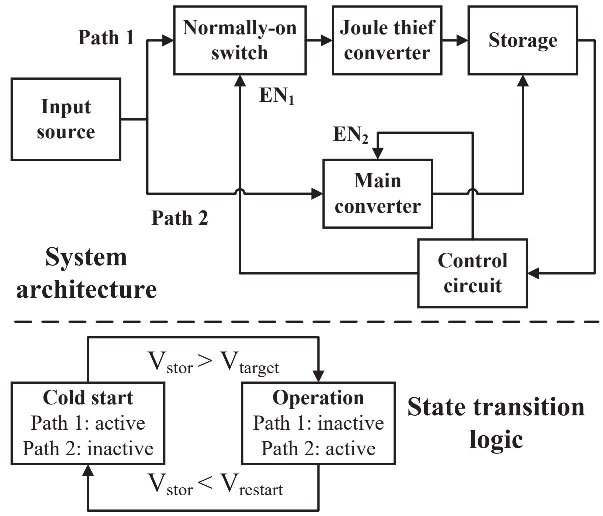

Fig. 22. System architecture for practical implementation: two-stage power management with state transition logic. The control circuit monitors the storage voltage and manages the transition between the cold-start Joule Thief (Path 1) and the high-efficiency main converter (Path 2).

standby, and minimal component count, enabling lower start-up acceleration than BQ25570. The modest efficiency and unregulated output are acceptable [9, 15], since the primary objective is reaching the main converter's threshold voltage—failure to do so yields zero system efficiency.

This positioning suggests a two-stage architecture, as illustrated in Fig. 22. Path 1 connects the source through a normally-on switch to the Joule Thief, charging the storage to the main PMIC threshold. Path 2 then takes over for high-efficiency conversion. State transitions follow hysteresis control: during Cold Start ($V_{stor} < V_{target}$), only Path 1 is active. Once V_{stor} exceeds V_{target}, the switch disables Path 1 to eliminate leakage. If V_{stor} drops below $V_{restart}$, the system reverts to Cold Start, with hysteresis preventing rapid toggling.

B. Model Accuracy, Limitations and Efficiency Analysis

For the BJT-based converter, the analytical model predicts V_{out} within $10\,\%$ of experimental results across the tested source-voltage and load conditions, as shown in Fig. 11. For the L_1 and a sweeps in Fig. 12, deviations between theory and simulation reach $\sim 20\,\%$ at extreme turns ratios, primarily due to two factors. First, the Ebers-Moll model assumes a constant β_F, but actual BJTs experience reduced gain at high currents due to the high-injection effect [27]. To compensate, we use $\beta_F = 100$, calibrated from measurements and the datasheet h_{FE} range (100–400) [24]. In contrast, the vendor SPICE model uses BF ≈ 360, yielding higher simulated V_{out}, as evidenced by the systematic overestimation (hollow markers vs. solid markers) in Fig. 11. Second, the simulation uses $k = 0.99$ while the model assumes ideal coupling $k = 1$, a difference amplified at extreme a.

For the NMOS-based converter, the model similarly predicts V_{out} within $10\,\%$ across the voltage, load, and source-resistance sweeps, as shown in Fig. 14. The sensitivity analysis in

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Table V confirms prediction errors within $\pm 7\%$ across sweeps of k, W_n/L_n, and A_d. The V_{out} error increases up to -5.1% as k decreases, because finite k reduces mutual inductance $M = k\sqrt{L_1 L_2}$ and thus feedback voltage. The NMOS aspect ratio sweep captures parasitic capacitance effects: since $C_{gs} \propto W_n L_n$, larger devices slow switching transitions, causing the model to overestimate oscillating frequency by 1–2%. The model shows low sensitivity to diode area variations, as the constant-V_D approximation remains valid over the tested range.

A key modeling limitation arises from the idealized boundary condition $i_1(0) = 0$ used to derive t_{on}, t_{off}, and η. At very high R_L ($> 10\,\mathrm{k\Omega}$), a large V_{gs} swing during turn-on induces a negative initial current via magnetic coupling, violating this assumption. Table VI quantifies this effect: as R_L increases from $10\,\mathrm{k\Omega}$ to $50\,\mathrm{k\Omega}$, the simulated $i_1(0)$ grows from $-6\,\mu\mathrm{A}$ to $-31\,\mu\mathrm{A}$, and the V_{out} prediction error increases from -1.2% to -4.0%. Notably, the i_{peak} prediction remains accurate ($\sim 1\%$ error) throughout, because i_{peak} is determined by $\partial i_{ds}/\partial V_{ds} = 0$ as in Eq. (15)—a condition independent of initial current—whereas the $i_1(0) = 0$ assumption directly enters t_{on} via Eq. (16) and propagates to V_{out}. Consequently, the model overestimates η, explaining the discrepancy in Fig. 15 at the highest R_L values. Accurately capturing this effect would require precise modeling of the gate charging and discharging dynamics through the magnetic coupling, leading to coupled nonlinear differential equations that preclude closed-form solutions. This limitation underscores that the analytical framework and simulation are complementary: the framework suits rapid design-space exploration, while simulation addresses parasitic effects idealized in theory.

The $38.3\,\%$ efficiency plateau in Fig. 15 is dictated by two loss mechanisms. First, conduction loss is significant due to the high on-resistance of the ALD212900 at ultra-low gate drives: $R_{ds} \approx 5\,\mathrm{k\Omega}$ at $V_{gs} = 0\,\mathrm{V}$, dropping to $1.18\,\mathrm{k\Omega}$ at $V_{gs} = 0.1\,\mathrm{V}$ [28], severely limiting efficiency at $V_{oc} = 300\,\mathrm{mV}$. Second, switching loss is exacerbated by the $R_{in}C_{gs}$ time constant ($C_{gs} \approx 30\,\mathrm{pF}$), which prolongs transitions [Fig. 13(b)] and directly increases voltage–current overlap loss. These constraints can be addressed through integration, as $\eta \approx 50\%$ is demonstrated in Table V and 74% in [17]. Thus, the discrete prototype primarily validates the model, while the topology offers a higher performance ceiling.

C. Prototype, Rectifier and Threshold Trade-off

MEMS-EMEHs exhibit significant diversity: Table I shows output voltages from $140\,\mathrm{mV}$ to $1.7\,\mathrm{V}$ and accelerations from $0.06\,\mathrm{g}$ to $7\,\mathrm{g}$, but all share high coil resistance (100–1500 Ω). Our prototype ($V_{oc} = 0.82\,\mathrm{V}$, $R_c = 275\,\Omega$) falls within these ranges and serves as an electrical emulator. This equivalence is grounded in the electromechanical model in Fig. 7: at resonance, the parallel L–C branch yields infinite impedance, causing mechanical contributions to cancel and leaving only resistive terms [26]. This explains why phase crosses zero at resonance ($53.9\,\mathrm{Hz}$ in Fig. 17) and why source characteristics depend on coil resistance rather than physical scale. Circuit compatibility with other MEMS-EMEHs is demonstrated in Fig. 21, where [5] ($R_c = 450\,\Omega$) reaches steady-state despite higher coil resistance than our prototype.

The impedance transformation through rectification, where optimal load shifts from $300\,\Omega$ (direct) to $2\,\mathrm{k\Omega}$ (rectified), underscores the need for explicit R_{in} modeling. This shift arises because the diode forward voltage drop acts as equivalent series resistance. The bridge rectifier here serves as proof-of-concept, and future implementations can employ active diodes [29] or cross-coupled rectifiers [30] to minimize loss. These active topologies achieve lower effective forward voltage drop (tens of mV), bringing a smaller optimal load.

All experimental results in Section IV-B were obtained at $54\,\mathrm{Hz}$, matching the harvester's resonance. This is representative of practical deployment, where EMEH resonance is designed to coincide with the dominant vibration frequency for maximum power extraction. Our prototype has a mechanical quality factor $Q = \sqrt{k_m m}/d_m \approx 27.6$ (3-dB bandwidth: $f_r/Q \approx 2\,\mathrm{Hz}$). As shown in Fig. 16, an 8% frequency offset (from $54\,\mathrm{Hz}$ to $50\,\mathrm{Hz}$) reduces the open-circuit voltage from $0.8\,\mathrm{V}$ to $\sim 0.2\,\mathrm{V}$, well below the rectifier threshold. Although source impedance remains nearly resistive at off-resonance (phase $< 0.3^\circ$ in Fig. 17), the harvester cannot provide sufficient voltage for cold-start. Compensating for this voltage drop by increasing acceleration is impractical, as it may exceed the proof mass displacement limits. To address off-resonance operation, frequency tuning techniques (e.g., adjusting mechanical stiffness via feedback circuits) [2] can be employed to match the harvester's resonant frequency to the excitation source.

The ZTVNC versus LTVNC comparison reveals a fundamental trade-off predictable from the model. From Eq. (15), i_{peak} scales with β_n and $(a+1)V_{oc} - V_{th}$, with V_{out} rising monotonically with i_{peak}. Table III quantifies this: the ALD212900 has $V_{th} = 0\,\mathrm{V}$ and $\beta_n = 0.015\,\mathrm{A\,V^{-2}}$, while the SiUD412ED has $V_{th} = 0.35\,\mathrm{V}$ and $\beta_n = 0.9\,\mathrm{A\,V^{-2}}$—a $60\times$ difference. The SiUD412ED's higher β_n yields larger i_{peak} and thus higher V_{out} ($93\,\%$ vs. $43\,\%$ boost), but its higher V_{th} demands higher startup voltage. Conversely, the ALD212900 starts at a minimal startup voltage but its low β_n limits boost. This trade-off can now be evaluated analytically without iterative simulation—a capability absent in prior work [17].

D. Systematic Design Workflow

The analytical framework enables rapid, simulation-free optimization. For given requirements (V_{oc}, R_{in}, target V_{out}), the design proceeds as follows:

1. Select the Active Device Based on Startup Requirements. The V_{th} choice depends on application priority: Zero-V_{th} devices enable ultra-low startup but limit boost due to low β_n, while low-V_{th} devices provide higher boost but require higher startup. For cost-sensitive applications not requiring ultra-low startup, BJTs offer a viable alternative.

2. Set the Turns Ratio (a) and Inductance (L_1). Choose a to meet target V_{out}, guided by Eq. (13) (BJT) or Eq. (21) (NMOS). Since V_{out} is largely insensitive to L_1, the selection only needs to ensure that the saturation current exceeds i_{peak} and meets size limits—this allows higher a to compensate for limited on-chip inductance in integrated designs.

3. Determine Biasing and Verify Safe Operation. Configure biasing to keep the device within Safe Operating Area.

TABLE VII
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART START-UP CIRCUIT TOPOLOGIES

Literature	Year	Circuit Topology	Min. V_{oc} (mV)	R_{in} (Ω)	Min. P_{AVL} (μW)	Efficiency η^*	w/ Theory
This work	2025	TF-based + zero-V_{th} NMOS	50	300	2.08	38.3 %@300 mV	Yes
[31]	2024	Startup oscillator + charge-pump	200	4.3	2325	87.6 %@200 mV	Yes
[14]	2024	TF-based + zero-V_{th} NMOS	11	0.9–3.5	8.6–33.6	22 %@13 mV	No
[13]	2024	TF-based + JFET	18	1.5	54	74.5 %@60 mV	Yes
[15]	2017	Piezoelectric TF-based + zero-V_{th} NMOS	15	2.8	20.1	–	Yes
			21	1000	0.11	–	
[32]	2016	Schmitt-trigger + charge pump	70	40	30.6	58 %@80 mV	No
[17]	2014	TF-based + zero-V_{th} NMOS	21	400	0.28	74 %@1000 mV	No

* Efficiency is peak reported value at the noted input voltage.

For BJT, adjust R_b to tune V_{out}. Since V_{CE} and V_{ds} peak during turn-off at approximately $V_{out} + V_D$, verify this value plus a 10–20% margin remains below maximum device ratings.

The output capacitor C_L serves as filter and energy buffer, with value affecting charging time and ripple rather than efficiency (low-ESR type preferred). Standard values of 10–100 μF are typically suitable.

E. Comparison with State-of-the-Art Start-up Circuits

Table VII compares this work with state-of-the-art start-up circuits using available source power $P_{AVL} = V_{oc}^2/(4R_{in})$ as the metric—the maximum output power under impedance-matched conditions, enabling fair comparison across various voltage–impedance combinations. The NMOS implementation achieves $50\,\mathrm{mV}$ start-up with $300\,\Omega$ impedance ($P_{AVL} = 2.08\,\mu\mathrm{W}$)—a rare combination of high-impedance compatibility and ultra-low voltage operation.

Most prior work targets low-impedance sources ($R_{in} < 10\,\Omega$) [13, 14, 31, 32]. Although [14] shares the self-oscillating principle and zero-V_{th} NMOS, it uses a two-stage architecture with modified Armstrong oscillator and 1:16 transformer, targeting 1–3 Ω sources without analytical model.

For high-impedance applications, two notable implementations achieve superior P_{AVL} thresholds. Although [15] achieves $0.11\,\mu\mathrm{W}$ using an Armstrong oscillator with piezoelectric transformer, its Barkhausen-based model requires numerical solution, the transformer's narrow bandwidth and load-dependent resonance drift limit applicability, and no efficiency is reported. Additionally, [17] employs a Joule Thief in $0.13\,\mu\mathrm{m}$ CMOS, achieving $P_{AVL} = 0.28\,\mu\mathrm{W}$ and $74\,\%@1000\,\mathrm{mV}/400\,\Omega$, but provides no analytical framework for design. Our closed-form model enables simulation-free performance prediction and trade-off navigation without iterative simulation, while discrete validation verifies theory rather than targeting record performance. For instance, while [17] adopts a fixed turns ratio ($a = 1$), our framework identifies a as critical for voltage gain. Future implementations could leverage CMOS integration, active diodes, and optimized turns ratio for enhanced performance.

VI. CONCLUSION

This work addresses the cold-start bottleneck in MEMS electromagnetic energy harvesting by developing a closed-form analytical framework for self-oscillating Joule Thief converters. The main contributions are summarized as follows:

1) **Closed-Form Analytical Framework:** A foundational model for BJT converters using the Lambert W function was developed and extended to NMOS implementations with explicit high source impedance incorporation. This enables simulation-free design optimization.
2) **Ultra-Low Voltage, High-Impedance Operation:** Laboratory characterization demonstrated $50\,\mathrm{mV}$ start-up from $300\,\Omega$ source impedance with the NMOS implementation, where the predicted output voltage agrees with experimental measurements to within $10\,\%$ over the tested V_{oc} and R_{in} ranges.
3) **System-Level Start-Up Demonstration:** Integration with a $300\,\Omega$ MEMS-representative EMEH reveals a practical trade-off: zero-threshold NMOS starts at $0.35\,\mathrm{g}$ RMS acceleration but provides $43\,\%$ boost; low-threshold NMOS starts at $0.5\,\mathrm{g}$ but achieves $93\,\%$ boost—both outperforming TI BQ25570's $0.7\,\mathrm{g}$ startup requirement.

These establish the Joule Thief as an effective cold-start front-end for MEMS-EMEHs and other high-impedance, low-voltage sources in two-stage power management.

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